

GramIQ

AI Ledger Digitization System

Technical Design Document

Version 1.0 • July 2026

Project	GramIQ AI Document Intelligence
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Status	Draft — For Internal Review
Version	1.0
Date	July 2026

1. Problem Statement

India has over 120 million smallholder farming households. A significant portion of these farmers maintain their financial records in handwritten paper notebooks — commonly known as Bahi-Khata. These physical ledgers capture critical data about:

- Seed purchases and variety names
- Fertilizer and pesticide expenditure
- Labour and machinery payments
- Irrigation and fuel costs
- Crop sales, dairy income, and loan records
- Miscellaneous farm expenses

Despite being valuable first-party financial records, these notebooks suffer from four compounding problems:

Problem	Impact
Physical & Unstructured	Records cannot be searched, aggregated, or analysed digitally
Multilingual Handwriting	Entries mix Hindi, Marathi, and English across the same page
Inconsistent Format	Date placement, column layout, and abbreviations vary per farmer
No Digital Continuity	Insights from past seasons are unavailable when making new crop decisions

The consequence is a structural information gap. Farmers cannot compute cost of cultivation, profit margins, or crop-wise profitability from these records without manual effort. Agribusinesses and financial institutions are likewise unable to access verified farm-level data to design relevant products or offer credit.

The GramIQ AI Ledger Digitization System directly addresses this gap by converting handwritten notebook images into structured, machine-readable financial transactions using a multi-stage AI pipeline.

2. Proposed Solution

The proposed solution is an end-to-end AI Document Intelligence pipeline embedded within the GramIQ Android application. Rather than a simple OCR tool, the system is designed as a multi-stage intelligence stack that understands agricultural context, handles mixed-language handwriting, and produces structured financial records with high accuracy.

2.1 Core User Journey

- Farmer opens the GramIQ app and taps "Scan Bahi-Khata"
- Captures one or more notebook pages using the camera
- The system validates image quality and prompts a retake if needed
- AI pipeline processes the image: enhances, reads, understands, and structures transactions
- Only low-confidence entries are shown for farmer review
- Verified transactions are stored and immediately visible in the farm finance dashboard
- Analytics, profit summaries, and crop-wise reports are updated automatically

2.2 Key Design Principles

Accuracy First	Transaction accuracy above 95% after review is the primary target. The system errs toward human review rather than incorrect auto-confirmation.
Agriculture-Aware	A domain-specific Farm Knowledge Base maps local names, abbreviations, and regional product aliases to standard categories — going beyond generic OCR.
Minimal Friction	High-confidence transactions are auto-approved. Farmers only see a review screen for uncertain entries, keeping the experience fast.
Multilingual	Hindi, Marathi, and English are treated as equal inputs. Mixed-language handwriting on the same page is supported natively.
Offline Capable	Lightweight on-device OCR is planned for low-connectivity regions in the next phase, ensuring rural usability.

3. Technical Architecture Overview

The system is structured as a linear processing pipeline where each stage produces an artifact consumed by the next. Stages are independently deployable and can be replaced or upgraded without rebuilding the full pipeline.

3.1 High-Level Pipeline

The pipeline flows as follows across eight stages:

#	Stage	Output Artifact
1	Image Acquisition	Raw image file (JPG / PNG)
2	Image Enhancement	Cleaned, deskewed, normalised image
3	Layout Analysis	Region map with typed bounding boxes
4	OCR Engine	Text + confidence + bounding box per line
5	Transaction Segmentation	Raw transaction records (date + text + amount)
6	Farm Intelligence Engine (LLM)	Categorised, enriched transaction JSON
7	Validation Engine	Confidence-scored, deduplicated transactions
8	Review UI + Database	Verified structured records

3.2 Component Architecture

Components are deployed across three tiers:

Mobile Tier (Flutter)

- Camera capture module with image quality pre-check
- Lightweight preview and user review screen
- Offline queue for uploads in low-connectivity areas

Backend Tier (FastAPI on Google Cloud)

- REST API gateway receiving image uploads
- Celery + Redis task queue for async pipeline execution
- Orchestration layer calling each pipeline stage sequentially
- Farm Knowledge Base microservice

AI / ML Tier

- OpenCV service for image enhancement
- PaddleOCR (primary) + TrOCR (fallback) for handwriting recognition
- PP-Structure or LayoutLMv3 for document layout understanding
- Gemini 2.5 Pro / GPT-5 for LLM-based parsing, categorisation, and enrichment
- Validation rules engine with arithmetic and duplicate checks

4. Deep Dive: Technical Decisions & Technologies

4.1 Image Enhancement — OpenCV

Handwritten notebook photos captured on mobile devices suffer from multiple quality issues that directly degrade OCR accuracy. OpenCV is chosen as the image pre-processing layer because it is battle-tested, runs efficiently on server CPUs, and provides every transformation needed:

- Deskew — corrects rotated pages using Hough line detection
- Denoising — applies non-local means denoising (cv2.fastNlMeansDenoising)
- Shadow removal — normalises uneven illumination with morphological operations
- Adaptive thresholding — converts grayscale to binary using local contrast (cv2.adaptiveThreshold)
- Perspective correction — flattens warped or angled notebook captures using homography

A quality gate is applied before enhancement. If the image falls below minimum thresholds for blur (Laplacian variance), resolution (minimum 300×400 px), or coverage, the app returns a prompt asking the farmer to retake the photo. This prevents propagating poor inputs into the expensive AI stages.

4.2 Layout Analysis — PP-Structure / LayoutLMv3

Farmer notebooks do not follow a uniform layout. Some use columns, others list items line-by-line, and many mix narrative text with tabular amounts. A dedicated layout analysis step is required to correctly identify transaction regions before OCR is applied.

Two models are recommended:

Model	Strength	When to Use
PP-Structure	Fast, lightweight, good for tabular/column layouts	Default for structured notebooks
LayoutLMv3	Multimodal (text + image), better on freeform pages	Fallback for unstructured entries

The output is a region map that tags each bounding box as one of: transaction, total/subtotal, header, or margin-note. Only transaction regions are forwarded to OCR. Totals and subtotals are detected and excluded from transaction extraction, preventing double-counting.

4.3 OCR Engine — PaddleOCR + TrOCR

OCR for Indian farmer notebooks presents two challenges that generic OCR systems do not handle well: mixed-script handwriting (Hindi Devanagari + Marathi + English in a single line), and informal abbreviations for agricultural products.

Primary: PaddleOCR

- Supports Hindi and English out-of-the-box with pre-trained models
- Faster inference than transformer-based alternatives
- Outputs text, per-character confidence scores, and bounding boxes
- Horizontally and vertically oriented text detection

Fallback: TrOCR (Microsoft)

- Transformer-based encoder-decoder architecture
- Stronger on irregular, degraded, or cursive handwriting
- Invoked automatically when PaddleOCR confidence falls below 0.75

Both engines return a standardised JSON structure per text line:

```
{ "text": "मजुरी", "confidence": 0.98, "bbox": [x1, y1, x2, y2] }
```

4.4 Farm Intelligence Engine — LLM (Gemini 2.5 Pro / GPT-5)

The most differentiated component of the system is the Farm Intelligence Engine — an LLM-powered parsing layer that understands agricultural context, not just text. Raw OCR output is incomplete: the product name "DAP" does not self-evidently map to "Fertilizer > Di-Ammonium Phosphate" without domain knowledge.

Responsibilities of the LLM layer:

- Parse OCR text into structured date / description / amount triplets
- Categorise each transaction (Fertilizer, Pesticide, Labour, Machinery, Sales, etc.)
- Resolve local product names and abbreviations against the Farm Knowledge Base
- Infer the crop being cultivated from contextual clues in surrounding entries
- Identify and skip subtotal rows
- Return a structured JSON array — no free text

Model selection rationale:

Model	Strength	Consideration
Gemini 2.5 Pro	Native multimodal, strong multilingual, GCP integration	Preferred — aligns with GramIQ GCP stack

Model	Strength	Consideration
GPT-5	Superior instruction following, robust JSON mode	Fallback / A-B evaluation partner

The system prompt provided to the LLM embeds the Farm Knowledge Base inline for the relevant crop and region, giving the model access to local spelling variants, product-to-category mappings, and amount sanity ranges. This retrieval-augmented approach keeps the model grounded without requiring fine-tuning.

4.5 Farm Knowledge Base

A domain-specific knowledge repository that maps unstructured agricultural references to canonical structured data. It is the backbone that distinguishes this system from a generic document digitisation tool.

Contents:

- Fertilizer names, aliases, and manufacturer brands (DAP → Di-Ammonium Phosphate → Fertilizer)
- Pesticide names, active ingredients, and subcategory (Monocrotophos → Insecticide)
- Seed varieties per crop and region
- Labour category names in Hindi and Marathi
- Common vendor names and aggregation aliases
- Expected amount ranges per category (sanity check bounds)
- Seasonal crop calendars per district in Maharashtra

The knowledge base is maintained as a structured database table (Product) and served via a FastAPI microservice. It is updated continuously as new product names and regional variants are encountered through farmer interactions on the broader GramIQ platform.

4.6 Validation Engine

Before any transaction reaches the review screen, it passes through a rule-based validation layer:

- Arithmetic validation — running totals on the page must reconcile with extracted amounts
- Date validation — dates are normalised and checked for plausibility (not in future, within current season)
- Amount validation — amounts checked against expected ranges from the Farm Knowledge Base
- Duplicate detection — hash-based comparison against recently uploaded pages for the same farmer
- Missing field detection — mandatory fields (date, amount, category) are flagged if absent
- Confidence scoring — each transaction receives a composite score (High / Medium / Low) based on OCR confidence and validation results

Only Low-confidence transactions are surfaced in the farmer review screen. High and Medium confidence transactions are auto-committed to the database, minimising the review burden.

4.7 Technology Stack Summary

Layer	Technology	Rationale
Mobile App	Flutter	Cross-platform; existing GramIQ codebase
API Gateway	FastAPI (Python)	High-performance async; native ML library support
Task Queue	Celery + Redis	Async pipeline execution; retries and prioritisation
Image Processing	OpenCV	Mature, efficient, comprehensive transform library
Layout Analysis	PP-Structure / LayoutLMv3	Purpose-built document layout models
OCR (Primary)	PaddleOCR	Multilingual handwriting; fast inference
OCR (Fallback)	TrOCR	Stronger on degraded / cursive handwriting
LLM / Intelligence	Gemini 2.5 Pro / GPT-5	Agricultural NLP, JSON mode, RAG-compatible
Database	PostgreSQL	Relational integrity for financial records
Object Storage	AWS S3	Durable, encrypted image archive
Cloud Platform	Google Cloud Platform	Existing GCP billing; Gemini API integration
Monitoring	Prometheus + Grafana	Pipeline latency and accuracy dashboards

5. API Design

Four REST endpoints expose the pipeline to the mobile application and future integrations:

POST /api/v1/notebooks/upload

Accepts a multipart image upload. Validates format and file size. Returns a notebook_id for subsequent calls.

```
Response: { "notebook_id": "uuid", "status": "Uploaded" }
```

POST /api/v1/notebooks/process

Triggers the async pipeline for a given notebook_id. Stages run in the Celery queue. Status is polled via GET.

```
Response: { "notebook_id": "uuid", "status": "Processing" }
```

GET /api/v1/notebooks/{id}/transactions

Returns all extracted transactions for the notebook, including confidence scores and verification status. The mobile app uses this to populate the review screen.

POST /api/v1/transactions/verify

Accepts farmer edits and approvals from the review screen. Updates transaction records to verified = true. Triggers downstream analytics recalculation.

6. Database Design

Four core tables store the digitisation artefacts and support downstream analytics:

6.1 Notebook

Field	Type	Description
id	UUID	Primary key
farmer_id	UUID	Foreign key to farmer profile
image_url	String	Signed S3 URL to original image
upload_time	Timestamp	UTC timestamp of upload
status	Enum	Uploaded / Processing / Review / Complete / Failed

6.2 Transaction

Field	Type	Description
id	UUID	Primary key
notebook_id	UUID	Parent notebook
transaction_date	Date	Normalised date
description	Text	Raw OCR description
category	String	e.g. Fertilizer, Labour, Sale
subcategory	String	e.g. Insecticide, Irrigation
crop	String	Inferred or farmer-confirmed crop
amount	Decimal	Transaction amount in INR
confidence	Float	Composite confidence 0.0–1.0
verified	Boolean	True after farmer review

7. Performance Targets & Security

7.1 Performance Targets

Metric	Target	Notes
Image Upload	< 3 seconds	Mobile to S3 via signed URL
OCR Processing	< 5 seconds	Per page
AI Parsing (LLM)	< 8 seconds	Per page; cached for repeated products
End-to-End Pipeline	< 15 seconds	Upload to review-ready
Transaction Accuracy	> 95%	After human review confirmation
OCR Character Accuracy	> 90%	Across Hindi, Marathi, English
System Availability	99.9% uptime	Monthly SLA

7.2 Security Controls

- HTTPS enforced on all API endpoints — no plaintext data transmission
- JWT-based authentication for all API calls from the mobile app
- Signed upload URLs — images upload directly to S3 without passing through application servers
- Encrypted image storage at rest on S3 (AES-256)
- Role-based access control (RBAC) — farmers see only their own records; agribusiness users see aggregated intelligence only
- Audit logging — all transactions record which pipeline stage created or modified them
- GDPR-aligned data handling — consistent with GramIQ's existing compliance framework

8. Expected Outcomes

Successful deployment of the AI Ledger Digitization System is projected to produce measurable outcomes across three dimensions:

8.1 Farmer-Level Outcomes

- Farmers who previously tracked finances only in paper notebooks gain instant digital visibility into crop-wise profit and loss
- Cost of cultivation per acre becomes computable without manual arithmetic
- Historical season data is preserved and searchable — no information lost when a notebook is damaged or lost
- Onboarding of existing farmers with years of Bahi-Khata records becomes possible, dramatically enriching GramIQ's data asset

8.2 Platform-Level Outcomes

- Significant uplift in Daily Active Users as digitisation provides immediate, tangible value to farmers
- First-party farm financial data at scale — the core proprietary asset required for GramIQ's agribusiness intelligence product
- Improved profitability prediction and crop planning models, trained on verified historical cost data
- Foundation for BFSI credit scoring — verified farm income and expense records can support farmer credit applications

8.3 Business-Level Outcomes

Metric	Projected Impact
Farmer Retention	Higher — digitisation creates switching cost and ongoing value
Data Richness	Depth of farm records increases agribusiness intelligence product value
New Revenue Streams	BFSI credit data products become viable once financial records are digitised
Differentiation	No competitor offers AI-powered Bahi-Khata digitisation in the Indian agritech market

9. Future Roadmap

The current TDD defines Phase 1: server-side AI pipeline. The following phases are planned:

Phase 2 — Voice Corrections

Allow farmers to correct transactions by voice: "Diesel amount is ₹900, not ₹700." The AI voice agent converts speech to a structured correction and updates the record.

Phase 3 — Multi-page Continuity

Automatically detect that a new upload is a continuation of a previously processed notebook and merge transactions into a single chronological ledger.

Phase 4 — On-Device OCR

Deploy a lightweight quantised OCR model directly on the Android app for farmers in low-connectivity areas. Transactions are extracted offline and synced when connectivity is restored.

Phase 5 — Proprietary Farm Document Model

Fine-tune a vision-language model specifically on GramIQ's growing dataset of farmer notebooks. This model will outperform generic models on agricultural handwriting, regional scripts, and product name resolution — creating a defensible AI asset.

Phase 6 — Financial Intelligence Layer

Automatically generate monthly P&L statements, crop-wise profitability reports, input cost benchmarks, and a financial health score — turning raw notebook data into actionable farm management intelligence.

10. Conclusion

The GramIQ AI Ledger Digitization System is a purpose-built, agriculture-domain AI pipeline that solves a problem no general-purpose document intelligence tool can address: reliably converting multilingual, handwritten Indian farm notebooks into structured financial data.

By combining computer vision (OpenCV), document layout analysis (PP-Structure / LayoutLMv3), multilingual handwriting OCR (PaddleOCR + TrOCR), large language model reasoning (Gemini 2.5 Pro / GPT-5), and a domain-specific Farm Knowledge Base, the system achieves transaction accuracy above 95% with minimal farmer effort.

The architecture is modular, scalable on Google Cloud Platform, and designed for continuous improvement. Each notebook processed enriches GramIQ's proprietary farm dataset — compounding the value of the Farm Intelligence Engine over time.

Beyond the immediate utility to individual farmers, the digitised records create the data foundation for GramIQ's higher-value products: agribusiness intelligence, BFSI credit scoring, and predictive crop planning. The Ledger Digitization System is therefore not only a farmer-facing feature — it is a strategic data infrastructure investment that strengthens GramIQ's competitive moat in the Indian agritech landscape.

Next Step	Engineering review of this TDD, assignment of stage owners, and sprint planning for Phase 1 pipeline implementation targeting Q3 2026.
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